

Week 2 – Progress Report

AI Agent for Supply Chain Operations
SE618-1 Logistics – Sirindhorn International Institute of Technology

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1 Activities Completed

1.1 Domain Research

We studied supply chain operations with a focus on understanding how data is structured across standard enterprise systems. We identified four key domains that form the backbone of a distribution company:

System	Scope	Key Data
SRM	Suppliers & Purchasing	Suppliers, Purchase Orders, Lead Times
WMS	Inventory	Products, Stock Levels, Warehouse Movements
OMS	Sales	Customers, Orders, Order Lines
TMS	Delivery	Shipments, Carriers, Tracking Events

Our research confirmed that **data silos** between these systems are a primary source of operational friction: answering a simple question like “*Why is order #123 delayed?*” can require manually querying three or four separate systems.

1.2 Tutor Meeting

We contacted and scheduled a first advisory meeting with our tutor, **Ajarn Warut** (SIIT), to present our topic and validate the project direction.

2 Roadmap

2.1 Phase 1: Supply Chain Modeling

Design and populate a simulated supply chain (PostgreSQL) with coherent test data across all four systems. Target scale: 5 suppliers, 30 products, 50 customers, ~500 orders over 3 months, 2 warehouses.

2.2 Phase 2: Agent Development (Two Tracks)

We will explore two complementary approaches to building the AI agent:

Track	Stack	Description
A: Custom Agent	Go, tRPC-Agent-Go	Build a fully custom agent with control over reasoning, tool routing, and orchestration. Our team is actively contributing to the open-source tRPC-Agent-Go framework.
B: MCP Integration	Go, MCP	Expose supply chain data via MCP servers (Model Context Protocol) and leverage existing agentic platforms (Claude, Claude Code) to query it.

Both tracks target the same core capability: enabling an AI agent to perform **read-only, cross-system reasoning**, autonomously navigating multiple data sources to diagnose problems, retrieve information, and surface actionable insights.

3 Next Steps

Begin an in-depth exploration of supply chain operations, studying how each system (SRM, WMS, OMS, TMS) functions individually, how data flows between them, and where the key interdependencies and pain points arise. This foundational understanding will guide the design of both the data model and the agent tooling.